

Total No. of Questions : 10]

SEAT No. :

P2143

[Total No. of Pages : 2

[5254]-539

B. E. (Mechanical)

PRODUCT DESIGN & DEVELOPMENT (Theory)
(2012 Pattern) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 70

nstructions to the candidates:

- 1) *All questions are compulsory.*
- 2) *Figure to the right indicate full marks.*

Q1) Discuss various factors that affect concept selection. Explain puzh's chart with suitable example. **[10]**

OR

- Q2)** a) Explain difference between product design & development. **[4]**
b) Explain different types of customer needs. **[6]**

Q3) Discuss product verification & product validation with suitable example. **[10]**

OR

Q4) Discuss How mission statement & technical questioning helps to design better products. Elaborate in detail. **[10]**

- Q5)** a) Explain various concerns in measurement phase during? tear down. **[8]**
b) What is cost control? Explain systematic procedure for calculation of cost. **[8]**

OR

- Q6)** a) Explain detailed procedure of Benchmarking. **[8]**
b) Explain any two types of product portfolio architectures in detail with suitable example. **[8]**

P.T.O.

- Q7)** a) Explain Guidelines of Design for manufacture. [8]
b) Discuss Guidelines of Design for welding. [8]

OR

- Q8)** a) Elaborate guidelines for Design for environment. [8]
b) What is life cycle assessment? Explain any one method in detail. [8]

- Q9)** a) Explain different phases of product life cycle with suitable example. [9]
b) Discuss elements of product life cycle management. [9]

OR

- Q10)** a) Explain different phases with corresponding technologies utilized in detail. [9]
b) Explain the importance of product data & product workflow in product life cycle management. [9]
